

Project: Subject: Designer: Date: **Eurocode 1**

Wind peak velocity pressure - United Kingdom and Ireland National Annex

Description:

Calculation of peak velocity pressure q_p depending on the basic wind velocity from National Annex wind map, and the site distance from shoreline and town boundary. The wind action on the structure (forces and pressures) can be derived from the peak velocity pressure. Calculation according to UK National Annex and Irish National Annex.

According to:

EN 1991-1-4:2005+A1:2010 Section 4 & UK National Annex to BS EN 1991-1-4:2005+A1:2010 & Irish National Annex to IS EN 1991-1-4:2005/NA:2013

SupportedNationalAnnexes:

United Kingdom National Annex, Irish National Annex

Input

Calculation method for the wind action	=	Exposure factor $C_e(z)$	▼
Fundamental basic wind velocity (before the altitude correction is applied)	$V_{b,map}$	= 22.64	m/s
Altitude of the site above mean sea level	A	= 32	m
Directional factor	C_{dir}	= 1.00 (direction 250 degrees)	▼
Upwind distance to shoreline along the examined wind direction	d_{Shore}	= 57.6	km
Upwind distance inside town terrain along the examined wind direction	d_{Town}	= 0	km
Height above ground at which peak velocity pressure is calculated (reference height)	Z	= 30	m
Displacement height for buildings in town	h_{dis}	= 0	m
Orography factor at reference height z	$c_0(z)$	= 1	
Season factor	C_{season}	= 1	

Results

Notes

1. The calculated peak velocity pressure $q_p(z)$ is the wind pressure corresponding to the peak wind velocity. In order to calculate the final wind actions on the structure (wind force or wind pressure) the appropriate force coefficient or pressure coefficient depending on the type of structure must be applied, as specified in EN1991-1-4 §5.2 and §5.3
2. The calculated wind action effects are characteristic values (unfactored). Appropriate load factors should be applied for the relevant design situation. For ULS verifications the partial load factor $\gamma_Q = 1.50$ is applicable for variable actions.
3. The authors of this website recommend the following pdf publication from SteelConstruction.info which provides in-depth analysis and worked examples regarding the calculation of wind loads according to EN1991-1-4 and the UK National Annex: [SteelConstruction.info publication SCI P394 - Wind Actions to BS EN1991-1-4 by A F Hughes](#).

Details

Input Data

- Calculation method for the wind action: = Exposure factor $C_e(z)$
- Fundamental basic wind velocity (before the altitude correction is applied): $v_{b, \text{map}} = 22.64 \text{ m/s}$
- Altitude of the site above mean sea level: $A = 32 \text{ m}$
- Directional factor: $c_{\text{dir}} = 1.00$ (direction 250 degrees)
- Upwind distance to shoreline along the examined wind direction: $d_{\text{Shore}} = 57.6 \text{ km}$
- Upwind distance inside town terrain along the examined wind direction: $d_{\text{Town}} = 0 \text{ km}$
- Height above ground at which peak velocity pressure is calculated (reference height): $z = 30 \text{ m}$
- Displacement height for buildings in town: $h_{\text{dis}} = 0 \text{ m}$
- Orography factor at reference height z : $c_0(z) = 1$
- Season factor: $c_{\text{season}} = 1$

Calculation of peak velocity pressure (UK National Annex to EN1991-1-4, Irish National Annex to EN1991-1-4)

Altitude correction factor

The effect of site altitude $A = 32 \text{ m}$ on the basic wind velocity is taken into account by the altitude correction factor c_{alt} that is determined from equations (NA.2a) and (NA.2b) of UK National Annex to EN1991-1-4 or Irish National Annex to EN1991-1-4. In this calculation the height above ground is considered conservatively as $0.6 \cdot z$:

$$c_{\text{alt}} = 1.028$$

Fundamental value of the basic wind velocity

The fundamental value of the basic wind velocity $v_{b,0}$ is defined in [EN1991-1-4 §4.2\(1\)P](#) as the characteristic 10 minutes mean wind velocity at 10 m above ground level. It is determined from eq. (NA.1) of UK National Annex to EN1991-1-4 or Irish National Annex to EN1991-1-4:

$$v_{b,0} = v_{b, \text{map}} \cdot c_{\text{alt}} = 22.64 \text{ m/s} \cdot 1.028 = 23.28 \text{ m/s}$$

where $v_{b, \text{map}} = 22.64 \text{ m/s}$ is the value of the fundamental basic wind velocity before the altitude correction is applied, that is given on the map in Figure NA.1 of UK National Annex to EN1991-1-4 or Irish National Annex to EN1991-1-4.

Basic wind velocity

The basic wind velocity v_b is defined in [EN1991-1-4 §4.2\(2\)P](#) as a function of the wind direction and time of year at 10 m above ground. It is calculated as:

$$v_b = c_{\text{dir}} \cdot c_{\text{season}} \cdot v_{b,0}$$

The directional factor c_{dir} takes into account the directional variation of the wind intensity. The appropriate value of the directional factor c_{dir} is provided in Table NA.1 of UK National Annex to EN1991-1-4 or Irish National Annex to EN1991-1-4 as a function of the wind direction. For the examined calculation the wind direction is considered as 250°, measured clockwise from North. The corresponding directional factor is obtained with linear interpolation from Table NA.1 as $c_{dir} = 1.00$.

The seasonal factor c_{season} represents the reduced wind intensity for structures that are loaded only for particular periods of the year. The appropriate value of the seasonal factor c_{season} is provided in Table NA.2 of UK National Annex to EN1991-1-4 or Irish National Annex to EN1991-1-4. For permanent structures that are loaded by the wind throughout the year $c_{season} = 1.0$. For the examined calculation the seasonal factor is considered as $c_{season} = 1.000$,

Therefore basic wind velocity v_b is calculated as:

$$v_b = c_{dir} \cdot c_{season} \cdot v_{b,0} = 1.00 \cdot 1.000 \cdot 23.28 \text{ m/s} = 23.28 \text{ m/s}$$

Orography factor

Where orography (e.g. hills, cliffs etc.) is significant its effect in the wind velocities should be taken into account using an orography factor $c_0(z)$ different than 1.0, as specified in [EN1994-1-1 §4.3.3](#). The recommended procedure in [EN1994-1-1 §4.3.3](#) for calculation of the orography factor $c_0(z)$ is described in [EN1994-1-1 §A.3](#).

In the following calculations the orography factor is considered as $c_0(z) = 1.000$.

Basic velocity pressure

The basic velocity pressure q_b is the pressure corresponding to the wind momentum determined at the basic wind velocity v_b . The basic velocity pressure is calculated according to the following fundamental relation, as specified in [EN1991-1-4 §4.5\(1\)](#):

$$q_b = (1/2) \cdot \rho \cdot v_b^2 = (1/2) \cdot 1.226 \text{ kg/m}^3 \cdot (23.28 \text{ m/s})^2 = 332 \text{ N/m}^2 = 0.332 \text{ kN/m}^2$$

where the density of the air is considered as $\rho = 1.226 \text{ kg/m}^3$ in accordance with the UK National Annex to [EN1991-1-4 §4.5\(1\)](#). Note that by definition $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$.

Terrain Category

The terrain category is considered as "Country" when the site is not inside town terrain or the upwind distance inside town terrain is $d_{Town} = 0.0 \text{ km} < 0.1 \text{ km}$.

Exposure factor for sites in Country terrain

The exposure factor for sites in Country terrain $c_e(z)$ is obtained from Figure NA.7 of UK National Annex to EN1991-1-4 or Irish National Annex to EN1991-1-4. The value of $c_e(z)$ depends on the effective height above ground $z - h_{dis}$ and the upwind distance to shoreline d_{shore} . For $z - h_{dis} = 30.000 \text{ m}$ and $d_{shore} = 57.6 \text{ km}$ the appropriate value is obtained from Figure NA.7 with double logarithmic interpolation as $c_e(z) = 3.094$.

Exposure correction factor for sites in Town terrain

For sites not inside Town terrain no correction to the exposure factor is applicable, i.e. $c_{e,T} = 1.000$.

Peak velocity pressure

The peak velocity pressure $q_p(z)$ at reference height z includes mean and short-term velocity fluctuations. For sites where either the orography is not significant ($c_0(z) = 1.0$) or $z \leq 50 \text{ m}$, the peak velocity pressure may be calculated in terms of the exposure factor $c_e(z) = 3.094$, as described in equations (NA.3a), (NA.3b) and (NA.4a) of UK National Annex to EN1991-1-4 or Irish National Annex to EN1991-1-4:

$$q_p(z) = c_e(z) \cdot c_{e,T} \cdot q_b \cdot [(c_0(z) + 0.6) / 1.6]^2 = 3.094 \cdot 1.000 \cdot 0.332 \text{ kN/m}^2 \cdot [(1.000 + 0.6) / 1.6]^2 = 1028 \text{ N/m}^2 \\ \Rightarrow q_p(z) = 1.028 \text{ kN/m}^2$$

Note that by definition $1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$.

Therefore the peak velocity pressure is calculated as $q_p(z) = 1.028 \text{ kN/m}^2$.

Calculation of wind forces and pressures on the structure

The wind actions on the structure (forces and pressures) depend on $q_p(z)$ as follows.

Wind pressures on surfaces

The wind pressure on surfaces are derived from the calculated value of $q_p(z) = 1.028 \text{ kN/m}^2$ by application of the appropriate pressure coefficient, as specified in [EN1991-1-4 §5.2](#).

For external surfaces the applicable wind pressure w_e is calculated as:

$$w_e = q_p(z) \cdot c_{pe}$$

where c_{pe} is the appropriate pressure coefficient for external pressure that is given in [EN1991-1-4 Section 7](#) depending on the type of structure. The appropriate reference height z for the external surface is given in [EN1991-1-4 Section 7](#) depending on the type of structure.

For internal surfaces the applicable wind pressure w_i is calculated as:

$$w_i = q_p(z_i) \cdot c_{pi}$$

where c_{pi} is the appropriate pressure coefficient for internal pressure that is given in [EN1991-1-4 Section 7](#) depending on the type of structure. The appropriate reference height z_i for the internal surface is given in [EN1991-1-4 Section 7](#) depending on the type of structure.

Total wind force on structure or structural element

The overall wind effect on the total structure or a particular structural element is expressed by the total wind force F_w that is estimated by application of the appropriate force coefficient, as specified in [EN1991-1-4 §5.3](#).

$$F_w = c_s c_d \cdot c_f \cdot q_p(z) \cdot A_{ref}$$

where:

- The structural factor $c_s c_d$ takes into account the structure size effects from the non-simultaneous occurrence of peak wind pressures on the surface and the dynamic effects from the effect of structural vibrations due to turbulence. The structural factor $c_s c_d$ is determined in accordance with [EN1991-1-4 Section 6](#) and the UK National Annex or Irish National Annex.
- The force coefficient c_f is given in [EN1991-1-4 Sections 7 and 8](#) depending on the type of structure or structural element.
- The wind reference area A_{ref} is given in [EN1991-1-4 Sections 7 and 8](#) depending on the type of structure or structural element.

